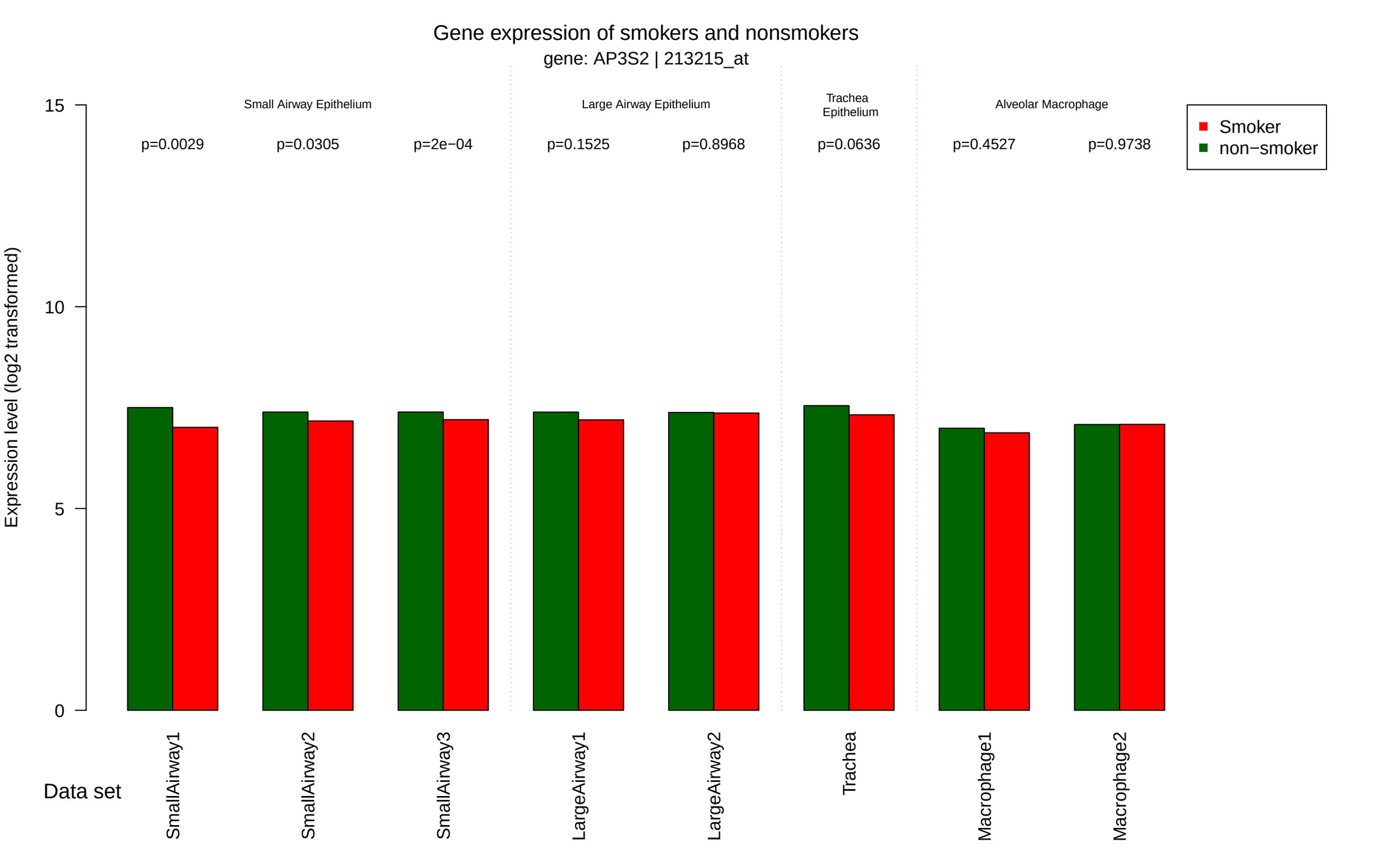
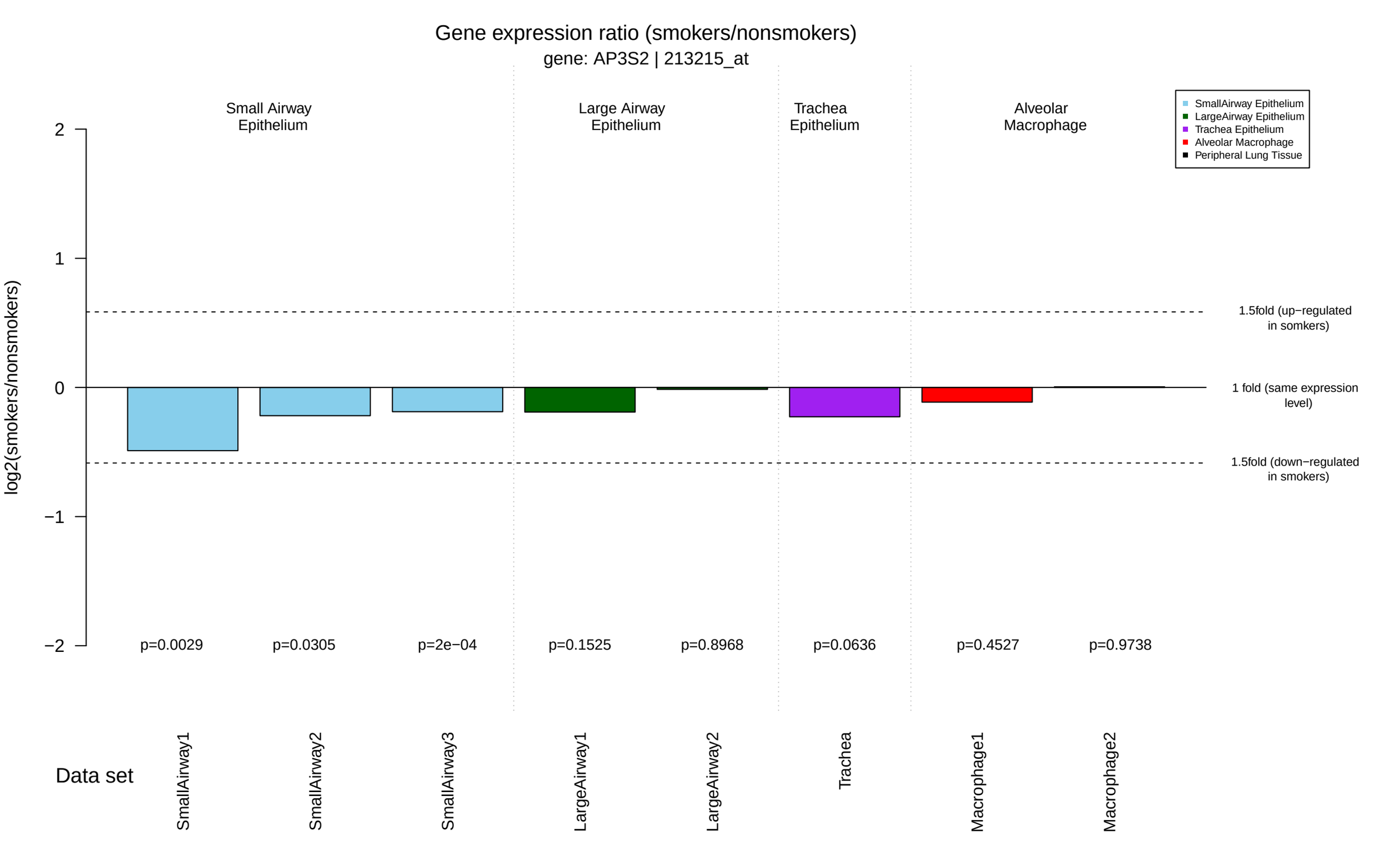
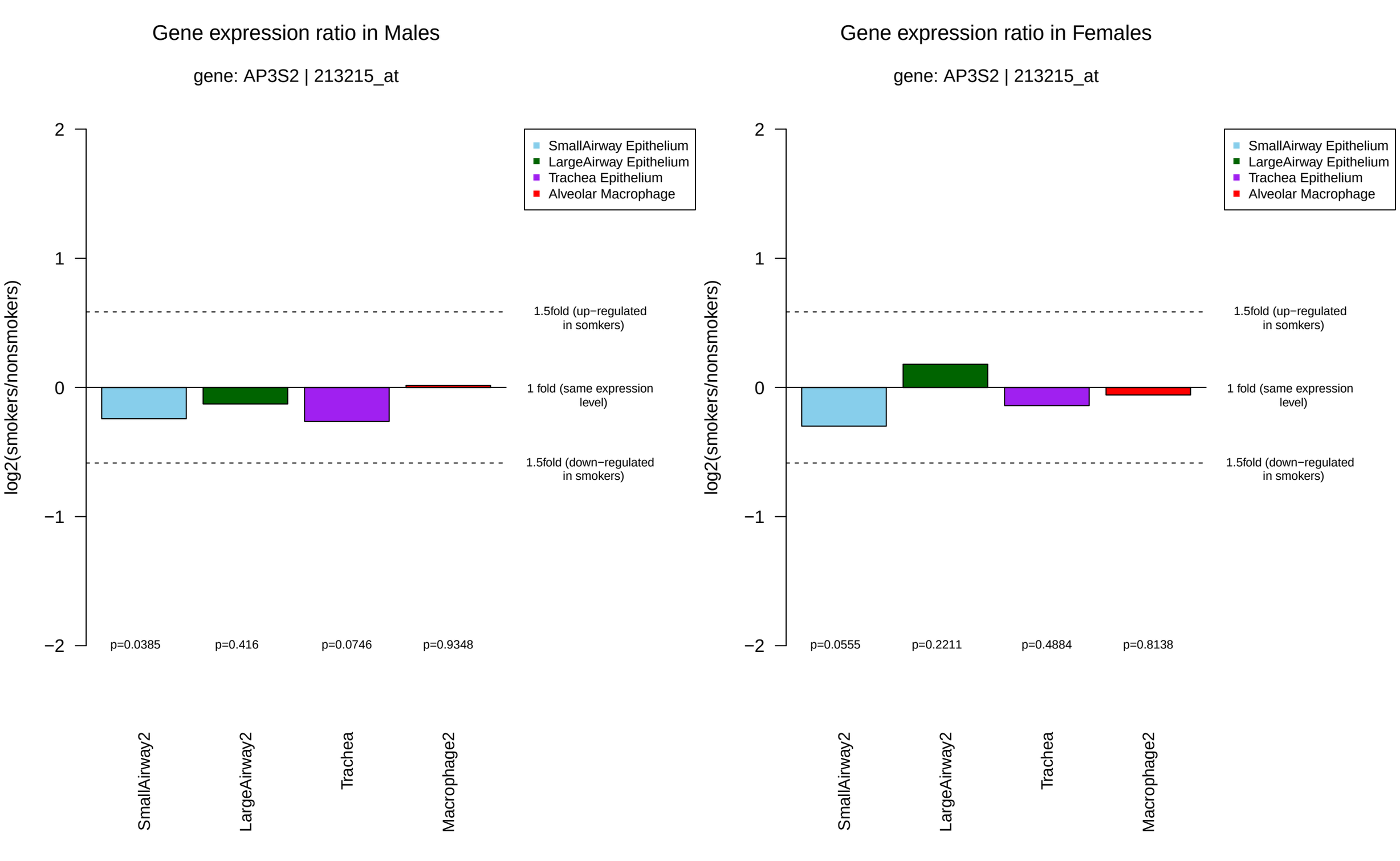
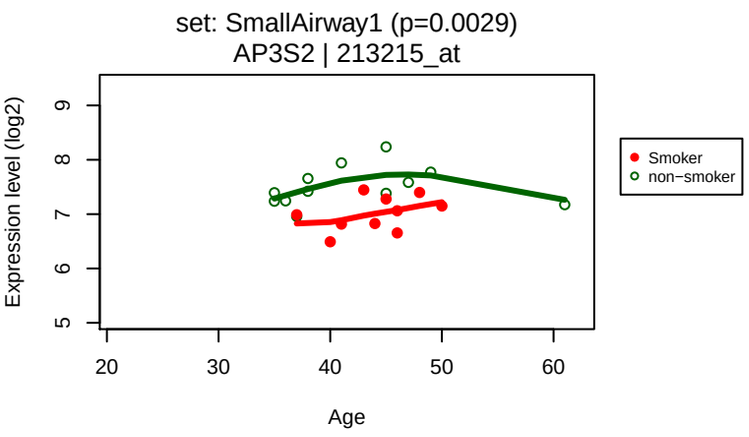
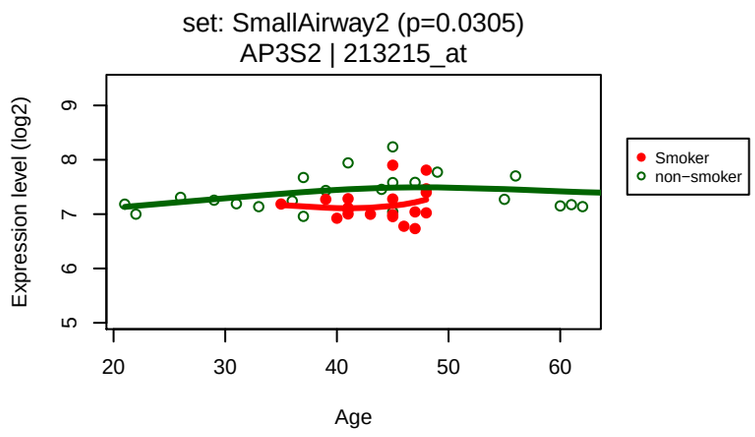
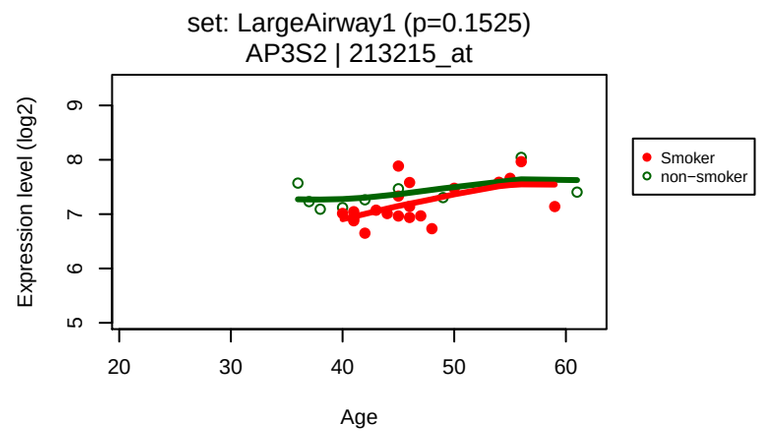

Gene expression of smokers and nonsmokers
gene: AP3S2 | 213215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0029
p=0.0305
p=2e−04
p=0.1525
p=0.8968
p=0.0636
p=0.4527
p=0.9738
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

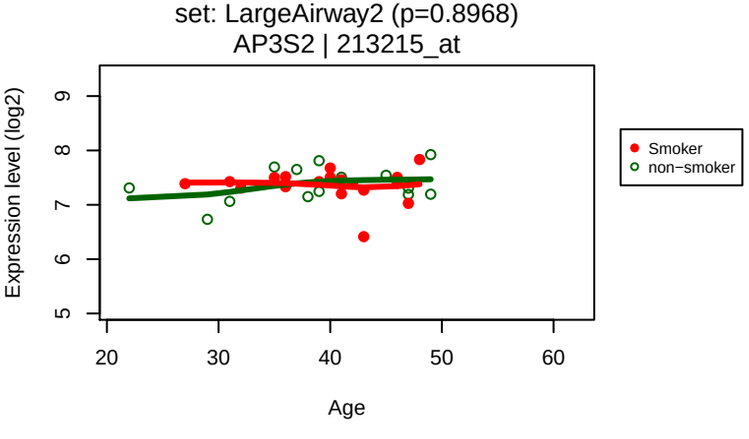

Gene expression ratio (smokers/nonsmokers)
gene: AP3S2 | 213215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0029
p=0.0305
p=2e−04
p=0.1525
p=0.8968
p=0.0636
p=0.4527
p=0.9738
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

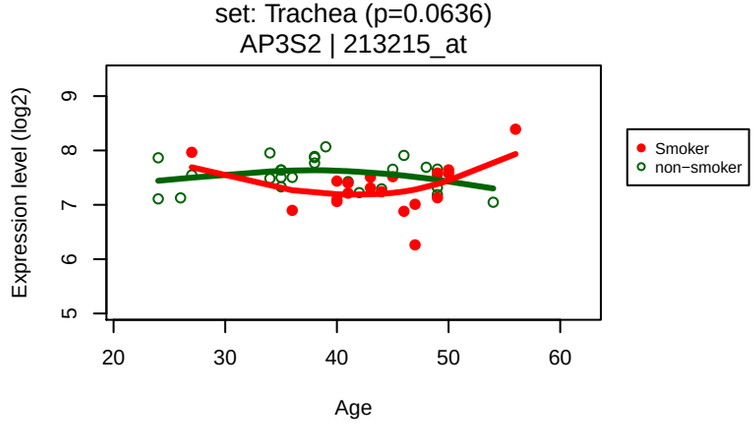

Gene expression ratio in Males
gene: AP3S2 | 213215_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0385
p=0.416
p=0.0746
p=0.9348
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP3S2 | 213215_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0555
p=0.2211
p=0.4884
p=0.8138
SmallAirway2
LargeAirway2
Trachea
Macrophage2

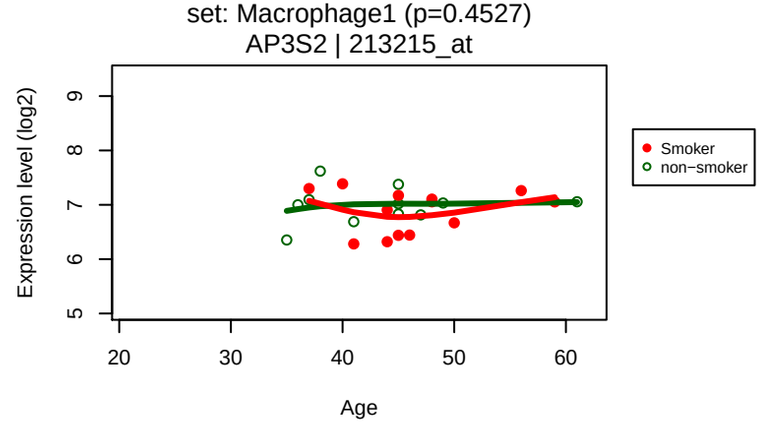

set: SmallAirway1 (p=0.0029)
AP3S2 | 213215_at
Expression level (log2)
Age
Smoker
non−smoker

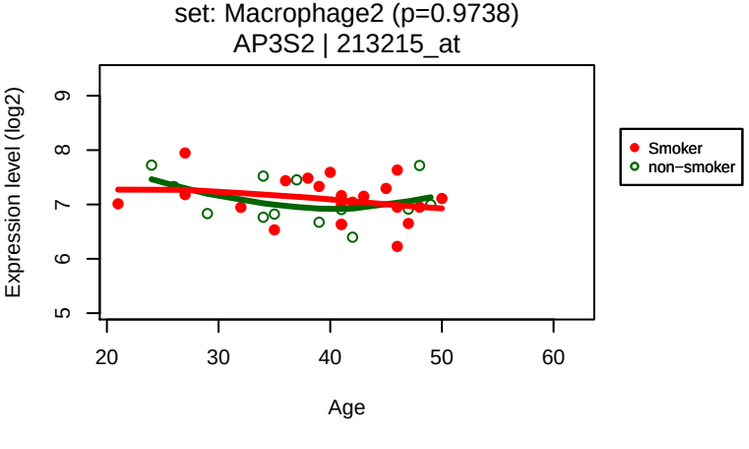

set: SmallAirway2 (p=0.0305)
AP3S2 | 213215_at
Expression level (log2)
Age
Smoker
non−smoker

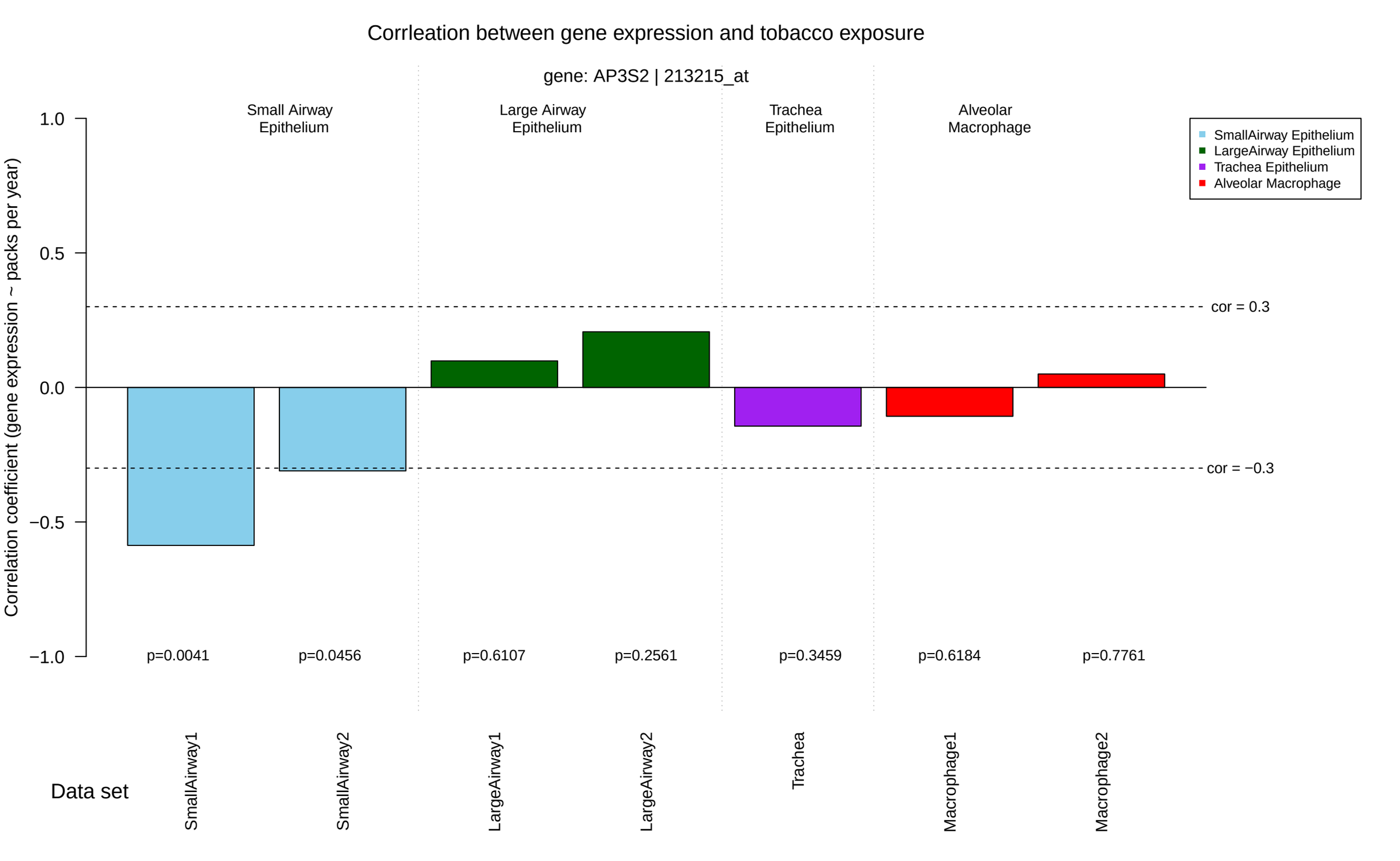

set: LargeAirway1 (p=0.1525)
AP3S2 | 213215_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8968)
AP3S2 | 213215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0636)
AP3S2 | 213215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4527)
AP3S2 | 213215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9738)
AP3S2 | 213215_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP3S2 | 213215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0041
p=0.0456
p=0.6107
p=0.2561
p=0.3459
p=0.6184
p=0.7761
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2